SRI LANKA NAVY — CIVIL ENGINEERING WING

NAVAL CIVIL ENGINEERING LOGISTICS & WAREHOUSING MANAGEMENT SYSTEM

CMSys Logistics (CE-Logistic-Panel-v3)

System Architecture, Polyglot Database Topology, 40-Model Schema Matrix, 22-Stage Procurement Pipeline & Central Warehouse Stores Inventory

System Platform:	CE-Logistic-Panel-v3 / CMSys Logistics
Core Technology Stack:	Next.js 15+ App Router, React 19, TypeScript, Tailwind CSS, Prisma ORM
Local Operational Cache:	SQLite 3 (WAL Mode, prisma/custom.db — 40 Relational Models)
MySQL Bridge Tier:	ncw_ps_db (127.0.0.1:3306 — 23 Relational Tables)
Cloud Firebase Topology:	cmsys-v2-1fb05 (Primary) + ncw-ps-operations + ce-admin-panel2025
Total Projects Tracked:	70 Capital Dockyard & Special Projects
Procurement Files Registered:	221 Purchase Requisition Files (LKR 84.5M+ Requisitions)
Central Warehouse Stock:	1860 Catalogued Stock Items across 11 Facility Zones
Naval Stakeholders Directory:	687 Registered Officers, Sailors & Area Store Keepers
Security & Governance:	Caddy HTTPS Reverse Proxy, OWASP Top 10 Protections, Role-Based Access Control

CLASSIFICATION: RESTRICTED // NAVAL STRATEGIC INTERNAL AUDIT
DIRECTORATE OF NAVAL CIVIL ENGINEERING // EASTERN NAVAL COMMAND

CMSys Logistics — Executive Architectural Summary & Multi-Tier Topology

1. Executive Architectural Summary & Scope

The Sri Lanka Navy Civil Engineering Logistics Management System (CE-Logistic-Panel-v3 / CMSys Logistics) operates as the mission-critical procurement, stores inventory, rate-contracting, and material distribution nexus for naval dockyards, base engineering workshops, and special community projects across all naval commands. It provides automated lifecycle management from initial Purchase Requisition (PR) vetting to quality assurance inspections (QC), store consignment take-on, and material issue to tactical civil works teams.

To guarantee operational resilience in demanding military dockyard environments, the system employs a modern polyglot, multi-tier architecture combining zero-latency local relational caching (SQLite 3 via Prisma ORM), enterprise relational interoperability (MySQL ncw_ps_db), high-availability cloud event streaming (Triple Firebase Realtime Database clusters), and edge network isolation (Caddy reverse proxy).

2. Multi-Tier System Topology & Infrastructure Matrix

TIER / SUBSYSTEM	TECHNOLOGY / PROTOCOL	PHYSICAL BINDING / ENDPOINT	OPERATIONAL ROLE
Presentation Tier	Next.js 15+ / React 19 / Tailwind	http://localhost:3000 (Internal)	Unified responsive dockyard web console, dark/emerald modes, PWA support
API & Microservices	Next.js Dynamic App Routes	src/app/api/* (30+ Endpoints)	REST controller routing for stores, procurement, stakeholders, sp-bridge
Local Operational Cache	SQLite 3 (WAL Engine)	file:./prisma/custom.db (1.68 MB)	High-speed local transaction store for 1,860 inventory items & 40 models
Data Access ORM	Prisma ORM v5/v6	prisma/schema.prisma	Automated TypeScript typing, schema migrations, relation enforcement
Relational DB Bridge	MySQL 8 / MariaDB	127.0.0.1:3306 (ncw_ps_db)	Federated link to CMSys V2 core tables: vendors, orders, jobs, people, users
Primary Firebase Cluster	Firebase RTDB (asia-southeast1)	cmsys-v2-1fb05-default-rtdb	Realtime cloud store sync under /logistics/ namespaces across commands
Tactical Operations Cluster	Firebase RTDB (asia-southeast1)	ncw-ps-operations-default-rtdb	Operational link to field job cards, daily allocations, site progress
Personnel Registry Cluster	Firebase RTDB (us-central1)	ce-admin-panel2025-default-rtdb	Read-only personnel stream: 661 active sailors, ranks, ratings, leaves
SP V1 Project Bridge	REST API & Local File Bridge	http://localhost/SP%20V1/api/api.php	Bi-directional absorption of Special Projects CSR into logistics files
Network Edge & Security	Caddy Web Server v2	Ports 80 & 443 (Reverse Proxy)	Automatic TLS certificate generation, rate limiting, origin protection

ARCHITECTURAL DESIGN INVARIANTS

- Offline-First Resilience: Local SQLite cache guarantees 100% store uptime during base network disconnects.
- Non-Colliding Namespaces: Cloud Firebase writes are isolated to /logistics/ to preserve CMSys V2 root structures.
- Cross-Tier Reconciliation: Nightly background workers reconcile SQLite stock balances against MySQL issue logs.
- Statutory Delegation Locking: Requisitions exceeding financial thresholds automatically lock pending authority approval.

3. Comprehensive Entity-Relationship Model Dictionary

The core business logic of CMSys Logistics is modeled via 40 strongly-typed Prisma entities spanning 8 distinct functional ERP modules: Projects, Procurement, Stores, Quality Assurance, Vendors, Governance, Security, and Configuration.

MODEL NAME	MODULE	FIELDS	SCOPE & FUNCTIONAL PURPOSE	PRIMARY RELATIONS
Project	Projects	28	Capital Dockyard & SP V1 CSR projects, vote allocations, EDC	ProjectFile, Allocation, Estimate, Earmarks
ProjectFile	Procurement	25	PR line items under Project, unit costs, present positions	Project, FileLog, FileApprovalEntry
FileApprovalEntry	Workflow	7	Extended audit trail of officer approvals, dispatches, authorizations	ProjectFile
FileLog	Workflow	7	Linear stage transition log tracking position history & comments	ProjectFile
Allocation	Finance	8	Fund allocation records (Commitment, Actual, Release) against projects	Project
ProcurementFile	Procurement	24	Consolidated procurement master entity with milestone tracking	ProcurementLog, QCInspection
ProcurementLog	Procurement	7	Stage transition history for ProcurementFile entries	ProcurementFile
Estimate	Estimates	6	Pre-construction cost estimates and BOQ package headers	Project, EstimateItem
EstimateItem	Estimates	22	Granular material and labour cost lines for project estimates	Estimate
EarmarkedItem	Inventory	11	Stock reservations for designated high-priority naval projects	Project, StoreItem
QCInspection	Quality Control	15	Material quality assurance inspection records, batch tests, pass/fail	ProcurementFile, StoreItem
Stakeholder	Personnel	19	Naval officers, sailors, subject clerks, appointment titles, area stores	AreaStoreRecord
Supplier	Vendors	13	Registered commercial vendors, contact details, business registration	SupplierContact, SupplierRating
SupplierContact	Vendors	9	Direct points of contact for vendors and emergency logistics	Supplier
SupplierNomination	Vendors	6	Tender board nominations and quotation submission tracking	ProcurementFile, Supplier
SupplierRating	Vendors	11	Vendor scorecards (delivery speed, compliance, quality metrics)	Supplier
ItemMaster	Catalog	11	Standardized naval item catalog definitions and specification codes	StoreItem, RateContract
StoreZone	Stores	10	Physical storage yards and warehouse zones (Zone A-H, Ready Use)	StoreItem
StoreItem	Stores	19	Live warehouse stock balances, min/max limits, unit costs, bin location	StoreZone, ItemMaster
RateContract	Pricing	17	Government approved rate contracts (RC), RC prices, VAT, brands	Supplier, ItemMaster
DiscontinuedItem	Stores	10	Obsolete, decommissioned or non-stock naval stores items	None
CollectionItem	Distribution	7	Items scheduled for collection by base engineering maintenance units	DailyCollection
PartSupplyOrder	Procurement	10	Split delivery orders and partial consignment tracking from suppliers	ProcurementFile
PartSupplyEntry	Procurement	15	Partial shipment arrival receipts and batch inspection records	PartSupplyOrder
DailyCollection	Distribution	19	Daily issue and store collection ledger for work order requisitions	CollectionItem, StoreZone
SplitOrderHeader	Distribution	7	Grouped consignment delivery headers for multi-location dispatch	SplitOrderSub
SplitOrderSub	Distribution	6	Sub-consignment dispatches to individual base workshops	SplitOrderHeader, SplitOrderEntry
SplitOrderEntry	Distribution	16	Line items within split order sub-consignments	SplitOrderSub, StoreItem
AreaStoreRecord	Stores	19	Area store sub-depots across naval commands (North, East, South)	Stakeholder
ProcessFlow	Workflow	8	Approval process flow definitions based on financial delegation tiers	ProcessStep, FinancialDelegation
ProcessStep	Workflow	17	Sequential stage definition within flow, responsible ranks/sailors	ProcessFlow
FinancialDelegation	Governance	6	Legal spending limits and approval authority tiers (DCO, CCFD, CCFO)	ProcessFlow
UserProfile	Security	11	User authentication profile, RBAC roles, service numbers, permissions	None
Vote	Finance	10	Naval budget Vote definitions, annual allocation, department head	Project
NavyListImport	Personnel	9	Import staging table for naval officer and sailor seniority lists	Stakeholder
StoreSetting	Config	11	Global stores parameters, low stock thresholds, currency, tax rates	None
NumberPrefix	Config	9	Document numbering prefix format rules (PR, RL, PO, MIV, GRN)	None
ThemeProfile	UI/UX	7	Custom UI color palettes (Naval Blue, Emerald, High Contrast, Dark)	None
ChangeLog	Audit	8	System-wide modification log capturing delta changes across all tables	None
SyncStatus	Telemetry	4	Cloud Firebase and MySQL synchronization health and timestamp tracker	None

4. The 22-Stage Naval Procurement Progression Pipeline

Every purchase requisition is governed by a strict sequential state machine that enforces multi-stage naval vetting, tender board compliance, budget certification, quality control, and store taking-on-charge:

PHASE	PIPELINE STAGE PROGRESSION	OPERATIONAL GOVERNANCE & CHECKS
Stage 01-04 (Inception)	PR Raised -> GS Approval -> Budget Check -> Tender Called	Requisition generated, verified against annual vote allocation, tender invited
Stage 05-08 (Evaluation)	Quotation Received -> BEC Evaluation -> AOE Approval -> PO Placed	Commercial bids evaluated by Bid Evaluation Committee (BEC), approved by AOE
Stage 09-12 (Consignment)	Order Placed -> Part Supply -> Items Dispatched -> In Transit	Purchase Order transmitted, supplier shipment tracking, consignment dispatch
Stage 13-16 (Receipt & QA)	Items Received -> QC Pending -> QC Passed / Rejected -> Awaiting Payment	Stores arrival inspection, technical sample verification, QA certificate issued
Stage 17-22 (Disbursement)	Payment Done -> Stores Take-on -> Bin Card Allocated -> Closed / Archived	Financial settlement, inventory charge ledger credit, official case closure.

5. Statutory Financial Delegations & Authority Limits

Procurement workflows automatically branch based on statutory financial thresholds defined under Sri Lanka Navy Procurement Regulations:

AUTHORITY LEVEL	DELEGATION LABEL	MAXIMUM VALUE (LKR)	PROCUREMENT METHOD	MANDATORY DOCUMENTATION
Commanding Officer (CO)	Direct Minor Works	LKR 100,000.00	Direct Local Purchase	Requisition Slip & 3 Quotations
Command Civil Engineer (CCE)	Routine Maintenance	LKR 500,000.00	Local Tender Board	BEC Report & Rate Comparison
Chief Civil Engineer (CCED)	Major Base Infrastructure	LKR 2,000,000.00	Departmental Tender Board	Full Specification Sheet & BOQ
Director General Operations (DGO)	Naval Strategic Works	LKR 10,000,000.00	Ministry Tender Board (MTB)	Technical Board Report & Vote Concurrence
Commander of the Navy (CN)	Capital Projects (Votes)	Unlimited (> LKR 10M)	Cabinet Appointed Tender Board	National Procurement Commission Filing

AUTOMATED PROCESS FLOW CONDITIONAL LOGIC

- Small Value Flow: Requisitions <= LKR 100,000 bypass BEC Evaluation and proceed directly from GS Approval to PO Placed.
- Standard Capital Flow: Requisitions > LKR 100,000 require 5-member BEC Technical Board sign-off and AOE Approval.
- Critical Fast-Track: Emergency Slipway / Dockyard repairs (NSH) trigger expedited 24-hour command concurrence.
- Automatic Rejection Loop: Items failing QC (QC Rejected) automatically halt invoice generation and issue supplier return notice.

6. Central Warehouse Stores Inventory Architecture

The stores inventory engine maintains 1,860 active line items valued across 11 designated physical facility zones. Inventory items are classified under specialized naval engineering trades (BMS, Plumbing, Metal, Paint, Electrical, Tools, Lubricant Oil, and General Stores). Each item tracks min/max safety stocks, verified suppliers, unit acquisition costs, and bin location identifiers.

ZONE IDENTIFIER	FACILITY TYPE	STOCK ITEM COUNT	TOTAL ZONE VALUATION (LKR)	PRIMARY STORAGE COMMODITIES
Zone A	Heavy Civil Materials Yard	412 Items	LKR 42,850,000.00	Cement, Tor Steel, Aggregates, Pre-cast Concrete, Quarry Dust
Zone B	Structural Steel Yard	285 Items	LKR 38,120,000.00	Universal Beams, MS Plates, Angle Irons, Galvanized Pipes
Zone C	Timber & Carpentry Yard	198 Items	LKR 14,650,000.00	Teak, Mahogany, Treated Plywood, Formwork Planks
Zone D	Plumbing & Waterworks Depot	224 Items	LKR 12,300,000.00	uPVC Pipes, S-Lon Fittings, Gate Valves, Water Pumps
Zone E	Electrical & Substation Stores	186 Items	LKR 18,940,000.00	Armoured Cables, Distribution Boards, Circuit Breakers, LED
Zone F	Marine Paints & Chemicals	142 Items	LKR 9,450,000.00	Epoxy Primers, Anti-Fouling Coatings, Thinners, Bitumen
Zone G	Power Tools & Plant Spares	165 Items	LKR 15,200,000.00	Welding Inverters, Angle Grinders, Core Drills, Concrete
Zone H	Fasteners & Hardware Depot	128 Items	LKR 4,850,000.00	High-Tensile Bolts, Foundation Anchors, Wire Ropes, Shackles
Ready-Use Stores	Daily Job Dispatch Center	54 Items	LKR 2,150,000.00	Fast-moving daily issue consumables for active
Balance Stores	Project Remainder Depot	38 Items	LKR 3,420,000.00	Retained surplus materials from completed capital naval
Project Stores	Dedicated Capital Site Yards	28 Items	LKR 6,950,000.00	Directly earmarked materials for Slipway Hangar (NSH) & Floating Dock

7. Inventory Valuation & Reorder Methodology

All warehouse transactions strictly follow the First-In, First-Out (FIFO) and Weighted Average Cost (WAC) models. Automatic inventory alerts trigger when stock falls below minStock thresholds, creating pending items in the Daily Collection ledger and triggering automated replenishment drafts under the master PR register.

8. Inter-System Bridging Architecture

CMSys Logistics does not operate in isolation; it functions as the central material and financial conduit connecting three external naval systems:

TARGET SYSTEM	INTERFACE PROTOCOL	DATA PAYLOAD / SCOPE	SYNCHRONIZATION MECHANISM
CMSys V2 Logistics Hub	Firebase REST & WebSocket	/logistics/projects, /logistics/storeitems	Realtime cloud push via firebase-
MySQL (ncw_ps_db)	mysql2/promise Pool (Port	work_orders, job_cards, locations,	Direct SQL querying of active work
SP V1 Projects Platform	Next.js API (/api/sp-bridge)	Special Projects CSR, BOQ bills,	Absorbs SP V1 projects and
Personnel (ce-admin-panel2025)	Firebase RTDB Live Stream	sailors, trades, official numbers, leaves	Realtime feed of authorized requisitioning officers and storekeepers

9. Network Security, Access Control (RBAC) & Hardening

In compliance with Sri Lanka Navy Information Security Directives and OWASP Top 10 guidelines, the application enforces:

SECURITY IMPLEMENTATION PROTOCOLS

- SQL Injection Protection: All database interactions utilize Prisma type-safe queries or PDO prepared statements.
- XSS & Sanitization: Strict React JSX text encoding and Next.js automatic input escaping.
- Caddy HTTPS Reverse Proxy: Automatic SSL/TLS encryption, HTTP/2 multiplexing, and internal port 3000 isolation.
- Role-Based Access Control (RBAC): DGO / CCED (Full Approval), Storekeeper (Read/Write Inventory), Sailor (Read Catalog).
- Audit Logging: Complete change log (ChangeLog model — 26 tracked events) capturing user, IP, timestamp, and field deltas.

10. Spreadsheet Deliverable Cross-Reference

The complete raw dataset, 40-model schema matrix, 70 master projects, 221 PR files, 1,860 store items, 11 zones, and 687 stakeholders are compiled into the official multi-worksheet Excel workbook:

MASTER EXCEL SPREADSHEET LEDGER

- File: CMSys_Logistics_System_Architecture_and_Database_Ledger.xlsx
- Worksheets: 9 Distinct Worksheets (Architecture, 40 Models, Projects, PR Files, Stores, Zones, Stakeholders, Flows, RC)
- Size: 1.55 MB of verified, fully formatted operational data cross-referenced with this report.